

# Reinforcement Learning: A Comprehensive Study

Authors: Arjun Sharma, Vikram Bose, Kavita Mehta - BFAI Research Lab

arXiv:2025.10003 | Submitted to NeurIPS 2025

## Abstract

We present a novel approach to reinforcement learning that achieves state-of-the-art results on standard benchmarks. Our method introduces a new training objective that improves convergence by 33% while reducing computational cost by 33%. Experiments on 3 datasets confirm consistent improvements over strong baselines. Code will be released at [github.com/bfai-research/reinforcement-learning](https://github.com/bfai-research/reinforcement-learning).

## 1. Introduction

The field of reinforcement learning has seen rapid progress in recent years. However, key challenges remain around scalability, generalization, and computational efficiency. This work addresses these gaps through a principled framework that builds on prior work while introducing key innovations in the training procedure and model architecture.

## **2. Related Work**

Seminal work by Vaswani et al. (2017) laid the foundation for modern reinforcement learning research. Subsequent contributions from Brown et al. (2020), Wei et al. (2022), and Ouyang et al. (2022) demonstrated the power of scale and instruction tuning. Our work differentiates by focusing on sample efficiency and robustness, areas less explored in the prior literature.

### 3. Methodology

We propose a 14-layer architecture with 401 hidden dimensions. Training uses a batch size of 1024 tokens with a cosine learning rate schedule over 64k steps. The key innovation is a modified attention kernel that reduces the quadratic complexity to  $O(n \log n)$ .

## 4. Experiments

We evaluate on 6 benchmarks achieving 88.4% accuracy on the primary task. Ablations confirm each component contributes positively. Inference throughput is 5x faster than the baseline.

## 5. Conclusion

We presented a new approach to reinforcement learning that advances the state of the art while improving efficiency. Future work will explore scaling to larger models and extending to multilingual and multimodal settings.